Mar 19 8:49 AM 42%

bilal@bilal-HP-Laptop-15-fc0xxx: /

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:/$ pwd
/
(base) bilal@bilal-HP-Laptop-15-fc0xxx:/$ ls
bin   cdrom  dev  home  lib32  lost+found  mnt  proc  run   snap  swap.img  tmp  var
boot  debug  etc  lib   lib64  media       opt  root  sbin  srv   sys       usr
(base) bilal@bilal-HP-Laptop-15-fc0xxx:/$ ls -l
total 8388688
lrwxrwxrwx   1 root root          7 Apr  9  2025 bin -> usr/bin
drwxr-xr-x   4 root root       4096 Mar 18 13:50 boot
drwxr-xr-x   2 root root       4096 Jan  1  1970 cdrom
-rw-r-----   1 root root        620 Dec  6 11:44 debug
drwxr-xr-x  19 root root       5040 Mar 18 13:56 dev
drwxr-xr-x 153 root root      12288 Mar 18 14:01 etc
drwxr-xr-x   3 root root       4096 Mar 18 11:32 home
lrwxrwxrwx   1 root root          7 Apr  9  2025 lib -> usr/lib
lrwxrwxrwx   1 root root          9 Sep  9  2025 lib32 -> usr/lib32
lrwxrwxrwx   1 root root          9 Apr  9  2025 lib64 -> usr/lib64
drwx------   2 root root      16384 Aug  3  2025 lost+found
drwxr-xr-x   3 root root       4096 Sep  9  2025 media
drwxr-xr-x   2 root root       4096 Apr 15  2025 mnt
drwxr-xr-x   8 root root       4096 Mar  9 08:26 opt
dr-xr-xr-x 461 root root          0 Mar 18 13:56 proc
drwx------  10 root root       4096 Dec 11 11:58 root
drwxr-xr-x  47 root root       1120 Mar 19 08:35 run
lrwxrwxrwx   1 root root          8 Apr  9  2025 sbin -> usr/sbin
drwxr-xr-x  28 root root       4096 Dec  9 11:44 snap
drwxr-xr-x   2 root root       4096 Apr 15  2025 srv
-rw-------   1 root root 8589934592 Aug  3  2025 swap.img
dr-xr-xr-x  13 root root          0 Mar 18 13:56 sys
drwxrwxrwt  21 root root        500 Mar 19 08:48 tmp
drwxr-xr-x  14 root root       4096 Nov 17 16:54 usr
drwxr-xr-x  15 root root       4096 Nov 22 15:26 var
(base) bilal@bilal-HP-Laptop-15-fc0xxx:/$
```

Mar 19 8:50 AM 42%

bilal@bilal-HP-Laptop-15-fc0xxx: /

```
-rw-------   1 root root 8589934592 Aug  3  2025 swap.img
dr-xr-xr-x  13 root root          0 Mar 18 13:56 sys
drwxrwxrwt  21 root root        500 Mar 19 08:48 tmp
drwxr-xr-x  14 root root       4096 Nov 17 16:54 usr
drwxr-xr-x  15 root root       4096 Nov 22 15:26 var
(base) bilal@bilal-HP-Laptop-15-fc0xxx:/$ ls -al
total 8388696
drwxr-xr-x  20 root root       4096 Dec  6 11:44 .
drwxr-xr-x  20 root root       4096 Dec  6 11:44 ..
lrwxrwxrwx   1 root root          7 Apr  9  2025 bin -> usr/bin
drwxr-xr-x   4 root root       4096 Mar 18 13:50 boot
drwxr-xr-x   2 root root       4096 Jan  1  1970 cdrom
-rw-r-----   1 root root        620 Dec  6 11:44 debug
drwxr-xr-x  19 root root       5040 Mar 18 13:56 dev
drwxr-xr-x 153 root root      12288 Mar 18 14:01 etc
drwxr-xr-x   3 root root       4096 Mar 18 11:32 home
lrwxrwxrwx   1 root root          7 Apr  9  2025 lib -> usr/lib
lrwxrwxrwx   1 root root          9 Sep  9  2025 lib32 -> usr/lib32
lrwxrwxrwx   1 root root          9 Apr  9  2025 lib64 -> usr/lib64
drwx------   2 root root      16384 Aug  3  2025 lost+found
drwxr-xr-x   3 root root       4096 Sep  9  2025 media
drwxr-xr-x   2 root root       4096 Apr 15  2025 mnt
drwxr-xr-x   8 root root       4096 Mar  9 08:26 opt
dr-xr-xr-x 462 root root          0 Mar 18 13:56 proc
drwx------  10 root root       4096 Dec 11 11:58 root
drwxr-xr-x  47 root root       1120 Mar 19 08:35 run
lrwxrwxrwx   1 root root          8 Apr  9  2025 sbin -> usr/sbin
drwxr-xr-x  28 root root       4096 Dec  9 11:44 snap
drwxr-xr-x   2 root root       4096 Apr 15  2025 srv
-rw-------   1 root root 8589934592 Aug  3  2025 swap.img
dr-xr-xr-x  13 root root          0 Mar 19 08:49 sys
drwxrwxrwt  21 root root        500 Mar 19 08:49 tmp
drwxr-xr-x  14 root root       4096 Nov 17 16:54 usr
drwxr-xr-x  15 root root       4096 Nov 22 15:26 var
(base) bilal@bilal-HP-Laptop-15-fc0xxx:/$
```

Mar 19 9:05 AM 36%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Downloads
~/Downloads

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ cd Downloads
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ cd ..
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ cd /home/bilal/Downloads
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ mkdir DevOps
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ rmdir DevOps
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ touch Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ touch Hello2.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ cp Hello.txt /home/bilal/practice
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ cd home/bilal/practice
bash: cd: home/bilal/practice: No such file or directory
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ cd /home/bilal/practice
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/practice$ ls
Hello.txt  git_practice
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/practice$ mv Hello.txt /home/bilal/Downloads
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/practice$ ls
git_practice
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/practice$ cd /home/bilal/Downloads
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ mv Hello.txt Rename.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ ls
 02999326-0ab9-41f0-8270-fcc3ec42e2e7.jpeg
 0937db62-4c77-4307-ae94-1fcad63917d6-md.jpeg
'0Q0ZD7AIZGM4ISJE4J5T9AF9FA.m3u8_in=62b5e97c219953839595f5191a2d83df__26fc10757934ead5bc2678c36cfc9c0e__1764198542__kp'
 1713893294574.jpeg
 1756658797835.jpeg
 1756659721982.jpeg
 1764921267275.jpeg
 1765207878095.jpeg
 1765903911491.jpeg
 1767522706121.jpeg
 '1768970765188 (1).jpeg'
 1768970765188-removebg-preview.png
 1768970765188.jpeg
 1768970826101-removebg-preview.png
 1768970826101.jpeg
 1772412839795.jpeg
```

Mar 19 9:11 AM 34%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Downloads
~/Downloads

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ rm Rename.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ mkdir deleteFolder
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ rm -r deleteFolder
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ touch Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ echo "Hello World"> Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ cat Hello.txt
Hello World
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ vim Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ more Hello.txt
Hello World
Tab 1
For the context, we need to export GA4 analytics data logs of a particular testing website and dump the data into bigquery. This data is concerned wit
h heatmaps. How it will gonna come in bigquery,how to process it further and the reason of doing so.

Analyze everything by section and then aggregate and group by all the information/ insights of heatmap.
No need to make all views, this is so much detail and we don't need it.


By section, we mean horizontal strips and components within each section, clear distinguish between component and section.


Only three things could have happened for each section (high level)
1. Users Who clicked here, got some value and move forward successfully
2. Users Who came here and bounced
3. Users Who navigated to some other section cz eventually they will bounce from some other section
In this way clear grouping in every section of three things that could have happened.

What is the best way to aggregate or do something with this data? Use all those amazing statistics which you can use like rage_clicks, miss clicks and
 where did they do and what did they do and what are the subevents that happened like hover etc.


Of all the users that followed this path we need to analyze their behaviour or intent.

This is one way to aggregate all this
```

Mar 19 9:15 AM 32%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Downloads
~/Downloads

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ less Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ less Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ head Hello.txt
Hello World
Tab 1
For the context, we need to export GA4 analytics data logs of a particular testing website and dump the data into bigquery. This data is concerned wit
h heatmaps. How it will gonna come in bigquery,how to process it further and the reason of doing so.

Analyze everything by section and then aggregate and group by all the information/ insights of heatmap.
No need to make all views, this is so much detail and we don't need it.


By section, we mean horizontal strips and components within each section, clear distinguish between component and section.

(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ tail Hello.txt

What is the best way to aggregate or do something with this data? Use all those amazing statistics which you can use like rage_clicks, miss clicks and
 where did they do and what did they do and what are the subevents that happened like hover etc.


Of all the users that followed this path we need to analyze their behaviour or intent.

This is one way to aggregate all this

What we are after is all those people who took some cta and moved forward from this page to navigate out of this page like terminated, (two ways for t
ermination 1. bounced or 2. took an action that navigated them out of that page.)

(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ tail -20 Hello.txt


By section, we mean horizontal strips and components within each section, clear distinguish between component and section.


Only three things could have happened for each section (high level)
1. Users Who clicked here, got some value and move forward successfully
```

Mar 19 9:20 AM 30%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Downloads — tail -f Hello.txt
~/Downloads

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ tail -f Hello.txt

What is the best way to aggregate or do something with this data? Use all those amazing statistics which you can use like rage_clicks, miss clicks and
 where did they do and what did they do and what are the subevents that happened like hover etc.


Of all the users that followed this path we need to analyze their behaviour or intent.

This is one way to aggregate all this

What we are after is all those people who took some cta and moved forward from this page to navigate out of this page like terminated, (two ways for t
ermination 1. bounced or 2. took an action that navigated them out of that page.)



^C
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ tail -f Hello.txt

What is the best way to aggregate or do something with this data? Use all those amazing statistics which you can use like rage_clicks, miss clicks and
 where did they do and what did they do and what are the subevents that happened like hover etc.


Of all the users that followed this path we need to analyze their behaviour or intent.

This is one way to aggregate all this

What we are after is all those people who took some cta and moved forward from this page to navigate out of this page like terminated, (two ways for t
ermination 1. bounced or 2. took an action that navigated them out of that page.)

tail: 'Hello.txt' has been replaced;  following new file
Hello World
Tab 1
For the context, we need to export GA4 analytics data logs of a particular testing website and dump the data into bigquery. This data is concerned wit
h heatmaps. How it will gonna come in bigquery,how to process it further and the reason of doing so.
```

Mar 19 9:27 AM 28%

bilal@bilal-HP-Laptop-15-fc0xxx: ~

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ whoami
bilal
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ hostname
bilal-HP-Laptop-15-fc0xxx
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ date
Thu Mar 19 09:22:38 PKT 2026
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ uptime
 09:22:54 up 19:26,  1 user,  load average: 0.46, 0.61, 0.68
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ uname -a
Linux bilal-HP-Laptop-15-fc0xxx 6.17.0-14-generic #14-Ubuntu SMP PREEMPT_DYNAMIC Fri Jan  9 17:01:16 UTC 2026 x86_64 GNU/Linux
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:            18Gi       7.6Gi       1.1Gi       437Mi        10Gi        10Gi
Swap:          8.0Gi          0B       8.0Gi
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.9G  2.5M  1.9G   1% /run
/dev/nvme0n1p2  468G  202G  242G  46% /
tmpfs           9.2G  317M  8.9G   4% /dev/shm
efivarfs        256K   55K  197K  22% /sys/firmware/efi/efivars
tmpfs           5.0M   16K  5.0M   1% /run/lock
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           9.2G   56M  9.2G   1% /tmp
/dev/nvme0n1p1  1.1G  6.3M  1.1G   1% /boot/efi
tmpfs           1.9G  116K  1.9G   1% /run/user/1000
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ top

top - 09:26:41 up 19:30,  1 user,  load average: 0.18, 0.43, 0.60
Tasks: 411 total,   1 running, 405 sleeping,   0 stopped,   5 zombie
%Cpu(s):  3.5 us,  1.3 sy,  0.0 ni, 94.9 id,  0.1 wa,  0.0 hi,  0.3 si,  0.0 st
MiB Mem :  18820.6 total,   1179.9 free,   7721.8 used,  10800.3 buff/cache
MiB Swap:   8192.0 total,   8192.0 free,      0.0 used.  11098.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
```

Mar 19 9:29 AM 27%

bilal@bilal-HP-Laptop-15-fc0xxx: ~ — htop

```
  0[||                        0.9%]  3[||                        0.9%]  6[||                        0.9%]  9[||                        0.9%]
  1[|                         0.5%]  4[||                        4.2%]  7[|                         0.5%] 10[||                        0.9%]
  2[||                        2.3%]  5[||                        1.8%]  8[||                        3.7%] 11[|                         0.5%]
Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||||||||6.38G/18.4G] Tasks: 196, 1246 thr, 212 kthr; 1 running
Swp[                                                          0K/8.00G] Load average: 0.50 0.45 0.58
                                                                        Uptime: 19:32:25

  Main   I/O
    PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
  28520 bilal      20   0 14440  7828  3640 R  6.0  0.0  0:04.23 htop
   3592 bilal      20   0 5374M  420M  174M S  1.8  2.2  7:02.63 /usr/bin/gnome-shell
  13218 bilal      20   0 1394G  198M     0 S  1.8  1.1  0:48.22 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=4706 --enable-cras
  13001 bilal      20   0 1394G  198M     0 S  1.4  1.1  0:44.54 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=4706 --enable-cras
   2392 mysql      20   0  958M  512M     0 S  0.9  2.7  0:32.61 /usr/sbin/mysqld
  20591 bilal      20   0 2201M  123M 91768 S  0.9  0.7  0:26.92 ptyxis --new-window
   2986 mongodb    20   0  549M  148M     0 S  0.5  0.8  0:31.01 /usr/bin/mongod --config /etc/mongod.conf
   4694 bilal      20   0 49.1G  770M  414M S  0.5  4.1  3:41.25 /opt/google/chrome/chrome
   4803 bilal      20   0 48.4G  152M     0 S  0.5  0.8  2:12.68 /opt/google/chrome/chrome --type=utility --utility-sub-type=network.mojom.NetworkSe
   5144 bilal      20   0 1394G  130M     0 S  0.5  0.7  0:01.68 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=4706 --enable-cras
   9462 root       20   0 1864M 47288     0 S  0.5  0.2  0:00.89 /usr/bin/containerd
      1 root       20   0 26720 16148  9812 S  0.0  0.1  0:08.00 /sbin/init splash splash
    422 root       19  -1 91560 46608 44728 S  0.0  0.2  0:01.88 /usr/lib/systemd/systemd-journald
    459 systemd-oo 20   0 18260  7240  6212 S  0.0  0.0  0:03.61 /usr/lib/systemd/systemd-oomd
    460 systemd-ti 20   0 91752  7564  6464 S  0.0  0.0  0:00.08 /usr/lib/systemd/systemd-timesyncd
    468 root       20   0 39068 12016  7484 S  0.0  0.1  0:01.40 /usr/lib/systemd/systemd-udevd
    469 systemd-re 20   0 23792 14420 11088 S  0.0  0.1  0:02.22 /usr/lib/systemd/systemd-resolved
    488 systemd-ti 20   0 91752  7564     0 S  0.0  0.0  0:00.00 /usr/lib/systemd/systemd-timesyncd
   1615 avahi      20   0  6744  4260  3844 S  0.0  0.0  0:00.37 avahi-daemon: running [bilal-HP-Laptop-15-fc0xxx.local]
   1619 root       20   0 13216  7052  6304 S  0.0  0.0  0:00.17 /usr/libexec/bluetooth/bluetoothd
   1622 messagebus 20   0 12108  8336  4112 S  0.0  0.0  0:03.24 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
   1672 polkitd    20   0  379M 11428  7736 S  0.0  0.1  0:00.67 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
   1695 root       20   0  305M  8476  7084 S  0.0  0.0  0:00.13 /usr/libexec/accounts-daemon
   1701 root       20   0  9856  3052  2804 S  0.0  0.0  0:00.03 /usr/sbin/cron -f -P
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

Mar 19 9:29 AM 27%

bilal@bilal-HP-Laptop-15-fc0xxx: ~

```
MiB Mem :  18820.6 total,   1104.6 free,   7794.9 used,  10789.6 buff/cache
MiB Swap:   8192.0 total,   8192.0 free,      0.0 used.  11025.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6195 bilal     20   0 4842852 193800  80680 S   5.0   1.0   4:18.58 language_server
  12999 bilal     20   0 1394.1g 203848 135424 S   2.6   1.1  19:22.70 chrome
    959 root     -51   0       0      0      0 S   1.3   0.0   2:34.50 irq/79-rtw89_pci
   2234 mysql     20   0  981784 525240  40832 S   1.3   2.7   0:53.82 mysqld
   2821 mongodb   20   0  562420 152468  76716 S   1.0   0.8   0:52.44 mongod
   5131 bilal     20   0 1392.5g 482712 150912 S   0.7   2.5   5:04.40 chrome
   5468 bilal     20   0 1394.0g 288012 173760 S   0.7   1.5   0:30.29 antigravity
   5855 bilal     20   0 1393.7g 245896  87304 S   0.7   1.3   0:47.58 antigravity
  28155 root      20   0       0      0      0 I   0.7   0.0   0:00.14 kworker/u48:2-writeback
     15 root      20   0       0      0      0 I   0.3   0.0   0:13.71 rcu_preempt
   3592 bilal     20   0 5507428 427860 178272 S   0.3   2.2   8:44.51 gnome-shell
   4694 bilal     20   0   49.1g 794508 429524 S   0.3   4.1  12:21.92 chrome
   4788 bilal     20   0   48.5g 155136 108348 S   0.3   0.8   2:22.69 chrome
   9454 root      20   0 1909748  47288  33264 S   0.3   0.2   0:07.98 containerd
  22004 bilal     20   0 1392.5g 423208 183680 S   0.3   2.2   2:47.54 chrome
  28460 bilal     20   0   13528   6056   3860 R   0.3   0.0   0:00.11 top
      1 root      20   0   26720  16148   9812 S   0.0   0.1   0:08.00 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root      20   0       0      0      0 I   0.0   0.0   0:00.46 kworker/0:1-events
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.52 ksoftirqd/0
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ htop
A signal 2 (Interrupt) was received, exiting without persisting settings to htoprc.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$
```

Mar 19 9:43 AM 30%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Downloads
~/Downloads

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ grep "horizontal" Hello.txt
By section, we mean horizontal strips and components within each section, clear distinguish between component and section.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ grep -i "horizontal" Hello.txt
By section, we mean horizontal strips and components within each section, clear distinguish between component and section.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ grep -i "Horizontal" Hello.txt
By section, we mean horizontal strips and components within each section, clear distinguish between component and section.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ grep "horizontal" Hello.txt
By section, we mean horizontal strips and components within each section, clear distinguish between component and section.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ grep "Horizontal" Hello.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ grep -i "Horizontal" Hello.txt
By section, we mean horizontal strips and components within each section, clear distinguish between component and section.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ find . -name ".txt"
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ find . -name "*.txt"
./3uToolsDownloads/cache/J417CFPYVW_info.txt
./3uToolsDownloads/cache/devices_table/devices_table.txt
./3uToolsDownloads/cache/cookie.txt
./3uToolsDownloads/cache/F17DW4MQ0DXR_info.txt
./🧠 Codet.txt
./unified-data-pipeline (2)/requirements.txt
./codet-datalog-server-bigquery/docs/api/endpoints-schema.txt
./codet-datalog-server-bigquery/backend/requirements.txt
./codet-datalog-server-bigquery/backend/endponits schema.txt
./Qwen_mermaid_20260223_8wu2wte5f.txt
./Hello2.txt
./nodes/all_modules_list.txt
./nodes/node_modules/jssha/src/license_header.es6.txt
./nodes/node_modules/jssha/src/license_header.es3.txt
./nodes/node_modules/import-in-the-middle/test/fixtures/esm-exports.txt
./nodes/node_modules/tslib/CopyrightNotice.txt
./nodes/node_modules/tslib/LICENSE.txt
./nodes/node_modules/typescript/ThirdPartyNoticeText.txt
./nodes/node_modules/typescript/LICENSE.txt
./nodes/node_modules/cliui/LICENSE.txt
./nodes/node_modules/cpu-features/deps/cpu_features/cmake/ci/sample/CMakeLists.txt
```

Mar 19 9:47 AM 34%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Downloads
~/Downloads

```
h heatmaps. How it will gonna come in bigquery,how to process it further and the reason of doing so.
Hello World
In this way clear grouping in every section of three things that could have happened.
No need to make all views, this is so much detail and we don't need it.
Of all the users that followed this path we need to analyze their behaviour or intent.
Only three things could have happened for each section (high level)
Tab 1
What is the best way to aggregate or do something with this data? Use all those amazing statistics which you can use like rage_clicks, miss clicks and
 where did they do and what did they do and what are the subevents that happened like hover etc.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ vim numbers
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ sort numbers
1
10
2
3
4
5
6
7
8
9
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ sort n numbers
sort: cannot read: n: No such file or directory
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$ sort -n numbers
1
2
3
4
5
6
7
8
9
10
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Downloads$
```

Mar 19 9:49 AM 37%

bilal@bilal-HP-Laptop-15-fc0xxx: ~

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ ping www.google.com
PING www.google.com (2001:4860:482a:7700::) 56 data bytes
64 bytes from 2001:4860:482a:7700::: icmp_seq=1 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=2 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=3 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=4 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=5 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=6 ttl=109 time=134 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=7 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=8 ttl=109 time=132 ms
64 bytes from 2001:4860:482a:7700::: icmp_seq=9 ttl=109 time=132 ms
^C
--- www.google.com ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8013ms
rtt min/avg/max/mdev = 131.634/132.047/133.622/0.595 ms
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ curl http://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{back
ground:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head
><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a h
ref="https://iana.org/domains/example">Learn more</a></p></div></body></html>
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ curl https://api.github.com
{"current_user_url":"https://api.github.com/user","current_user_authorizations_html_url":"https://github.com/settings/connections/applications{/client
_id}","authorizations_url":"https://api.github.com/authorizations","code_search_url":"https://api.github.com/search/code?q={query}{&page,per_page,sort
,order}","commit_search_url":"https://api.github.com/search/commits?q={query}{&page,per_page,sort,order}","emails_url":"https://api.github.com/user/em
ails","emojis_url":"https://api.github.com/emojis","events_url":"https://api.github.com/events","feeds_url":"https://api.github.com/feeds","followers_
url":"https://api.github.com/user/followers","following_url":"https://api.github.com/user/following{/target}","gists_url":"https://api.github.com/gist
s{/gist_id}","hub_url":"https://api.github.com/hub","issue_search_url":"https://api.github.com/search/issues?q={query}{&page,per_page,sort,order}","is
sues_url":"https://api.github.com/issues","keys_url":"https://api.github.com/user/keys","label_search_url":"https://api.github.com/search/labels?q={qu
ery}&repository_id={repository_id}{&page,per_page}","notifications_url":"https://api.github.com/notifications","organization_url":"https://api.github.
com/orgs/{org}","organization_repositories_url":"https://api.github.com/orgs/{org}/repos{?type,page,per_page,sort}","organization_teams_url":"https://
api.github.com/orgs/{org}/teams","public_gists_url":"https://api.github.com/gists/public","rate_limit_url":"https://api.github.com/rate_limit","reposi
tory_url":"https://api.github.com/repos/{owner}/{repo}","repository_search_url":"https://api.github.com/search/repositories?q={query}{&page,per_page,s
ort,order}","current_user_repositories_url":"https://api.github.com/user/repos{?type,page,per_page,sort}","starred_url":"https://api.github.com/user/s
tarred{/owner}{/repo}","starred_gists_url":"https://api.github.com/gists/starred","topic_search_url":"https://api.github.com/search/topics?q={query}{&
page,per_page}","user_url":"https://api.github.com/users/{user}","user_organizations_url":"https://api.github.com/user/orgs","user_repositories_url":"
```

Mar 19 9:49 AM 37%

bilal@bilal-HP-Laptop-15-fc0xxx: ~

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether c0:bf:be:e6:18:8c brd ff:ff:ff:ff:ff:ff
    altname wlp1s0
    altname wlxc0bfbee6188c
    inet 192.168.18.124/24 brd 192.168.18.255 scope global dynamic noprefixroute wlo1
       valid_lft 2748sec preferred_lft 2748sec
    inet6 2400:adcc:122:fe00::b/128 scope global dynamic noprefixroute
       valid_lft 81949sec preferred_lft 81949sec
    inet6 2400:adcc:122:fe00:a5db:2498:527:29d7/64 scope global temporary dynamic
       valid_lft 86205sec preferred_lft 81822sec
    inet6 2400:adcc:122:fe00:9870:46c2:b5db:2612/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86205sec preferred_lft 86205sec
    inet6 fe80::1225:ec83:368a:3edb/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 22:95:a0:aa:5c:47 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether 22:95:a0:aa:5c:47  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 42 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
```

Mar 19 10:03 AM 54%

bilal@bilal-HP-Laptop-15-fc0xxx: ~

```
udp6       0      0 :::54456                :::*                                    23805/python3
udp6       0      0 :::5353                 :::*                                    -
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$  ss -tulpn
Netid    State      Recv-Q     Send-Q                        Local Address:Port          Peer Address:Port    Process
udp      UNCONN     0          0                                127.0.0.54:53                0.0.0.0:*
udp      UNCONN     0          0                               127.0.0.53%lo:53              0.0.0.0:*
udp      UNCONN     0          0                                     0.0.0.0:33145           0.0.0.0:*        users:(("python3",pid=23805,fd=11))
udp      UNCONN     0          0                                     0.0.0.0:51100           0.0.0.0:*        users:(("python3",pid=23805,fd=8))
udp      UNCONN     0          0                                     0.0.0.0:51435           0.0.0.0:*
udp      UNCONN     0          0                                172.17.0.1:3702              0.0.0.0:*        users:(("python3",pid=23805,fd=12))
udp      UNCONN     0          0                            239.255.255.250:3702             0.0.0.0:*        users:(("python3",pid=23805,fd=10))
udp      UNCONN     0          0                             192.168.18.124:3702             0.0.0.0:*        users:(("python3",pid=23805,fd=9))
udp      UNCONN     0          0                            239.255.255.250:3702             0.0.0.0:*        users:(("python3",pid=23805,fd=7))
udp      UNCONN     0          0                               224.0.0.251:5353              0.0.0.0:*        users:(("chrome",pid=4788,fd=115))
udp      UNCONN     0          0                               224.0.0.251:5353              0.0.0.0:*        users:(("chrome",pid=4694,fd=119))
udp      UNCONN     0          0                               224.0.0.251:5353              0.0.0.0:*        users:(("chrome",pid=4788,fd=18))
udp      UNCONN     0          0                                     0.0.0.0:5353            0.0.0.0:*
udp      UNCONN     0          0                 [fe80::1225:ec83:368a:3edb]%wlo1:546            [::]:*
udp      UNCONN     0          0                                        [::]:35714           [::]:*
udp      UNCONN     0          0                 [fe80::1225:ec83:368a:3edb]%wlo1:3702           [::]:*        users:(("python3",pid=23805,fd=15))
udp      UNCONN     0          0                                 [ff02::c]%wlo1:3702           [::]:*        users:(("python3",pid=23805,fd=13))
udp      UNCONN     0          0                                             *:54456              *:*        users:(("python3",pid=23805,fd=14))
udp      UNCONN     0          0                                        [::]:5353            [::]:*
tcp      LISTEN     0          151                                  127.0.0.1:3306              0.0.0.0:*
tcp      LISTEN     0          4096                                127.0.0.54:53                0.0.0.0:*
tcp      LISTEN     0          70                                   127.0.0.1:33060             0.0.0.0:*
tcp      LISTEN     0          4096                              127.0.0.53%lo:53               0.0.0.0:*
tcp      LISTEN     0          4096                                 127.0.0.1:631               0.0.0.0:*
tcp      LISTEN     0          200                                  127.0.0.1:5432              0.0.0.0:*
tcp      LISTEN     0          4096                                 127.0.0.1:27017             0.0.0.0:*
tcp      LISTEN     0          4096                                     [::1]:631               [::]:*
tcp      LISTEN     0          511                                          *:80                  *:*
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~$ sudo ss -tulpn | grep :80
[sudo: authenticate] Password:
tcp   LISTEN 0      511                                  *:80                *:*    users:(("apache2",pid=2229,fd=4),("apache2",pid=2227,fd=4),("apach
e2",pid=2225,fd=4),("apache2",pid=2224,fd=4),("apache2",pid=2222,fd=4),("apache2",pid=2188,fd=4))
```

Mar 19 10:23 AM 55%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Desktop
~/Desktop

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Desktop$ zip schemas.zip schemas
  adding: schemas/ (stored 0%)
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Desktop$ unzip schemas.zip
Archive:  schemas.zip
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Desktop$
```

Mar 19 10:23 AM 55%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/Desktop
~/Desktop

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Desktop$ zip schemas.zip schemas
  adding: schemas/ (stored 0%)
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Desktop$ unzip schemas.zip
Archive:  schemas.zip
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/Desktop$ tar -cvf schemas.tar schemas
schemas/
schemas/supply-chain-procurement-management_schema.json
schemas/purchase-order-management_schema.json
schemas/lead-management_schema.json
schemas/policy-management_schema.json
schemas/legal-case-tracker_schema.json
schemas/emergency-triage-log_schema.json
schemas/incident-management_schema.json
schemas/business-expenses_schema.json
schemas/contract-management_schema.json
schemas/personal-finance-manager_schema.json
schemas/andon-calls_schema.json
schemas/beverage-sales-management_schema.json
schemas/office-guest-log_schema.json
schemas/order-kiosk_schema.json
schemas/sqcdp_schema.json
schemas/compliance-assessment-builder_schema.json
schemas/esg-management_schema.json
schemas/gemba-walks_schema.json
schemas/standard-operating-procedures_schema.json
schemas/sales-pipeline-management_schema.json
schemas/personal-health-log_schema.json
schemas/non-emergency-call-center_schema.json
schemas/healthcare-facility-management_schema.json
schemas/individual-medical-record_schema.json
schemas/risk-assessment-and-management_schema.json
schemas/lightweight-crm_schema.json
schemas/call-center-log_schema.json
schemas/health-inspection-reports_schema.json
schemas/electronic-health-record_schema.json
```

Mar 19 10:41 AM 49%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/devops-week1-git-lab
~/DevOps/week1/devops-week1-git-lab

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ https://github.com/bilalijaz786/devops-week1-git-lab.git
bash: https://github.com/bilalijaz786/devops-week1-git-lab.git: No such file or directory
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ git clone https://github.com/bilalijaz786/devops-week1-git-lab.git
Cloning into 'devops-week1-git-lab'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ ls
devops-week1-git-lab
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ cd devops-week1-git-lab
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ ls
README.md
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ echo "This is my first DevOps Git Lab." >>intro.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        intro.txt

nothing added to commit but untracked files present (use "git add" to track)
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git add .
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git commit -m "intro file is added"
[master 1eee9cb] intro file is added
 1 file changed, 1 insertion(+)
 create mode 100644 intro.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/bilalijaz786/devops-week1-git-lab.git'
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git push origin master
Username for 'https://github.com': bilalijaz82
Password for 'https://bilalijaz82@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 310 bytes | 310.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

Mar 19 10:44 AM 48%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/devops-week1-git-lab
~/DevOps/week1/devops-week1-git-lab

```
 create mode 100644 intro.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/bilalijaz786/devops-week1-git-lab.git'
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git push origin master
Username for 'https://github.com': bilalijaz82
Password for 'https://bilalijaz82@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 310 bytes | 310.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/bilalijaz786/devops-week1-git-lab.git
   d8b42f8..1eee9cb  master -> master
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git checkout -b feature/about-me
Switched to a new branch 'feature/about-me'
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ echo "My name is Muhammad Bilal Ijaz and I'm learning DevOps" >> about.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git add .
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$ git commit -m "Added about me section"
git push origin feature/about-me
[feature/about-me d15c3ca] Added about me section
 1 file changed, 1 insertion(+)
 create mode 100644 about.txt
Username for 'https://github.com': bilalijaz82
Password for 'https://bilalijaz82@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 365 bytes | 365.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/about-me' on GitHub by visiting:
remote:      https://github.com/bilalijaz786/devops-week1-git-lab/pull/new/feature/about-me
remote:
To https://github.com/bilalijaz786/devops-week1-git-lab.git
 * [new branch]      feature/about-me -> feature/about-me
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/devops-week1-git-lab$
```

Mar 19 10:48 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | New Tab | devops-week1-git-l

github.com/bilalijaz786/devops-week1-git-lab/tree/master?files=1

AWS Cloud Pra... | Upwork home... | BWT - Data En... | Home - Micros... | Console Home... | Introduction t... | All Bookmarks

bilalijaz786 / **devops-week1-git-lab**

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Files

master

Go to file t

- README.md
- about.txt
- intro.txt

devops-week1-git-lab /

Add file

bilalijaz786 Merge pull request #1 from bilalijaz786/feature/about-me d5a6a5b · 1 minute ago History

Name	Last commit message	Last commit date
README.md	Initial commit	13 minutes ago
about.txt	Added about me section	4 minutes ago
intro.txt	intro file is added	9 minutes ago

README.md

devops-week1-git-lab

Mar 19 10:49 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | New Tab | bilalijaz786/devop

github.com/bilalijaz786/devops-week1-git-lab

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bilalijaz786 / devops-week1-git-lab

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devops-week1-git-lab

Public

Pin | Watch 0 | Fork 0 | Star 0

master | 1 Branch | 0 Tags | Go to file t | Code

bilalijaz786 Merge pull request #1 from bilalijaz786/feature/about-me d5a6a5b · 2 minutes ago 4 Commits

README.md	Initial commit	14 minutes ago
about.txt	Added about me section	5 minutes ago
intro.txt	intro file is added	10 minutes ago

README

devops-week1-git-lab

About

No description, website, or topics provided.

Readme
Activity
0 stars
0 watching
0 forks

Releases

No releases published
Create a new release

Mar 19 10:49 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | New Tab | bilalijaz786/devop

github.com/bilalijaz786/devops-week1-git-lab/pulls?q=is%3Apr+is%3Aclosed

AWS Cloud Pra... | Upwork home... | BWT - Data En... | Home - Micros... | Console Home... | Introduction t... | All Bookmarks

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Label issues and pull requests for new contributors

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Dismiss

Filters | is:pr is:closed | Labels 9 | Milestones 0 | New pull request

Clear current search query, filters, and sorts

0 Open | 1 Closed | Author | Label | Projects | Milestones | Reviews | Assignee | Sort

Added about me section

#1 by bilalijaz786 was merged 2 minutes ago

ProTip! Add no:assignee to see everything that's not assigned.

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Mar 19 10:54 AM 45%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/team-devops-profile-feature
~/DevOps/week1/team-devops-profile-feature

bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/devops-week1-git-lab | bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/team-devops-profile-feature

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ git clone https://github.com/bilalijaz786/team-devops-profile-feature.git
Cloning into 'team-devops-profile-feature'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ ls
devops-week1-git-lab  team-devops-profile-feature
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ cd team-devops-profile-feature
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git checkout -b feature/user-profile
Switched to a new branch 'feature/user-profile'
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ echo "My name is Muhammad Bilal Ijaz , I am from peshawar ,Done my
bachelors in Software Engineering ,Currently learning DevOps" >> about-me.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$
```

Mar 19 11:04 AM 41%

bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/team-devops-profile-feature
~/DevOps/week1/team-devops-profile-feature

bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/devops-week1-git-lab | bilal@bilal-HP-Laptop-15-fc0xxx: ~/DevOps/week1/team-devops-profile-feature

```
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ git clone https://github.com/bilalijaz786/team-devops-profile-feature.git
Cloning into 'team-devops-profile-feature'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ ls
devops-week1-git-lab  team-devops-profile-feature
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1$ cd team-devops-profile-feature
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git checkout -b feature/user-profile
Switched to a new branch 'feature/user-profile'
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ echo "My name is Muhammad Bilal Ijaz , I am from peshawar ,Done my
bachelors in Software Engineering ,Currently learning DevOps" >> about-me.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git add .
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git commit -m "about-me file added"
[feature/user-profile 0351a9e] about-me file added
 1 file changed, 1 insertion(+)
 create mode 100644 about-me.txt
(base) bilal@bilal-HP-Laptop-15-fc0xxx:~/DevOps/week1/team-devops-profile-feature$ git push origin feature/user-profile
Username for 'https://github.com': bilalijaz82
Password for 'https://bilalijaz82@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 384 bytes | 384.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
```

Mar 19 11:04 AM 41%

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | Git Collaboration & | bilalijaz786/team-c

github.com/bilalijaz786/team-devops-profile-feature

AWS Cloud Pra... | Upwork home... | BWT - Data En... | Home - Micros... | Console Home... | Introduction t... | All Bookmarks

bilalijaz786 / team-devops-profile-feature

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

team-devops-profile-feature

Public

Pin | Watch 0 | Fork 0 | Star 0

feature/user-profile had recent pushes 17 seconds ago

Compare & pull request

master | 2 Branches | 0 Tags | Go to file t | + | Code

bilalijaz786 Initial commit 9009e19 · 14 minutes ago 1 Commit

README.md Initial commit 14 minutes ago

README

team-devops-profile-feature

https://github.com/bilalijaz786/team-devops-profile-feature/compare/feature/user-profile?expand=1

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Mar 19 11:04 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | Git Collaboration & | bilalijaz786/team-c

github.com/bilalijaz786/team-devops-profile-feature/compare/feature/user-profile?expand=1

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Create pull request

Remember, contributions to this repository should follow our GitHub Community Guidelines.

Helpful resources

GitHub Community Guidelines

1 commit | 1 file changed | 1 contributor

Commits on Mar 19, 2026

about-me file added

Bilal Ijaz committed 1 minute ago

0351a9e

Showing 1 changed file with **1 addition** and **0 deletions**.

Split | Unified

1 about-me.txt

```
@@ -0,0 +1 @@
+ My name is Muhammad Bilal Ijaz , I am from peshawar ,Done my bachelors in Software Engineering ,Currently learning DevOps
```

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Mar 19 11:04 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | Git Collaboration & | bilalijaz786/team-c

github.com/bilalijaz786/team-devops-profile-feature/compare/feature/user-profile?expand=1

AWS Cloud Pra... | Upwork home... | BWT - Data En... | Home - Micros... | Console Home... | Introduction t... | All Bookmarks

Add a title *

about-me file added

Add a description

Write | Preview

Added about-me.txt
- Introduced a short personal introduction
- This helps team members know who contributed
- No additional notes

Markdown is supported | Paste, drop, or click to add files

Create pull request

Remember, contributions to this repository should follow our GitHub Community Guidelines.

Reviewers
No reviews

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Use Closing keywords in the description to automatically close issues

Helpful resources
GitHub Community Guidelines

Mar 19 11:05 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | Git Collaboration & | about-me file adde

github.com/bilalijaz786/team-devops-profile-feature/pull/1

AWS Cloud Pra... | Upwork home... | BWT - Data En... | Home - Micros... | Console Home... | Introduction t... | All Bookmarks

Conversation 0 | Commits 1 | Checks 0 | Files changed 1

bilalijaz786 commented now — Owner

Added about-me.txt

- Introduced a short personal introduction
- This helps team members know who contributed
- No additional notes

about-me file added — 0351a9e

No conflicts with base branch
Merging can be performed automatically.

Merge pull request — You can also merge this with the command line. View command line instructions.

Still in progress? Convert to draft

Add a comment

Reviewers
No reviews
Still in progress? Convert to draft

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

Mar 19 11:05 AM

(19) WhatsApp | intrenship_cloudlle | Git Pull vs Fetch | Git Collaboration & | about-me file adde

github.com/bilalijaz786/team-devops-profile-feature/pull/1

AWS Cloud Pra... | Upwork home... | BWT - Data En... | Home - Micros... | Console Home... | Introduction t... | All Bookmarks

Merged

about-me file added #1

bilalijaz786 merged 1 commit into `master` from `feature/user-profile`

- Introduced a short personal introduction
- This helps team members know who contributed
- No additional notes

`about-me file added` `0351a9e`

bilalijaz786 merged commit **ee13abc** into `master` now

Revert

Pull request successfully merged and closed

You're all set — the `feature/user-profile` branch can be safely deleted.

Delete branch

Add a comment

Write | Preview

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

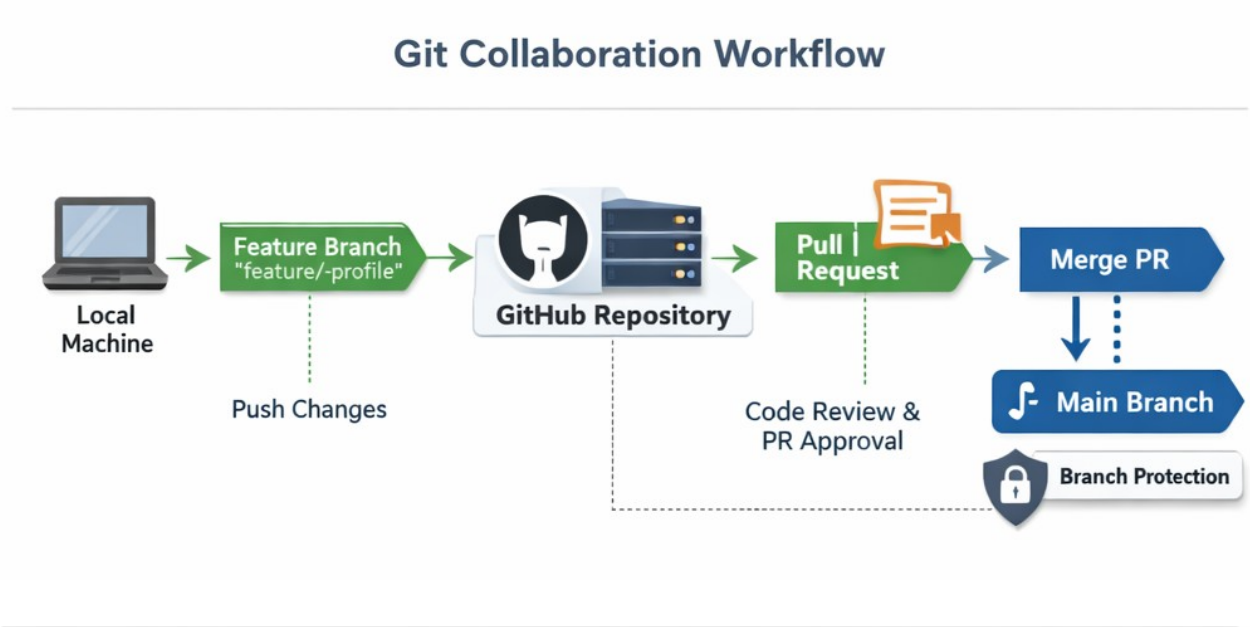
Successfully merging this pull request may close these issues.

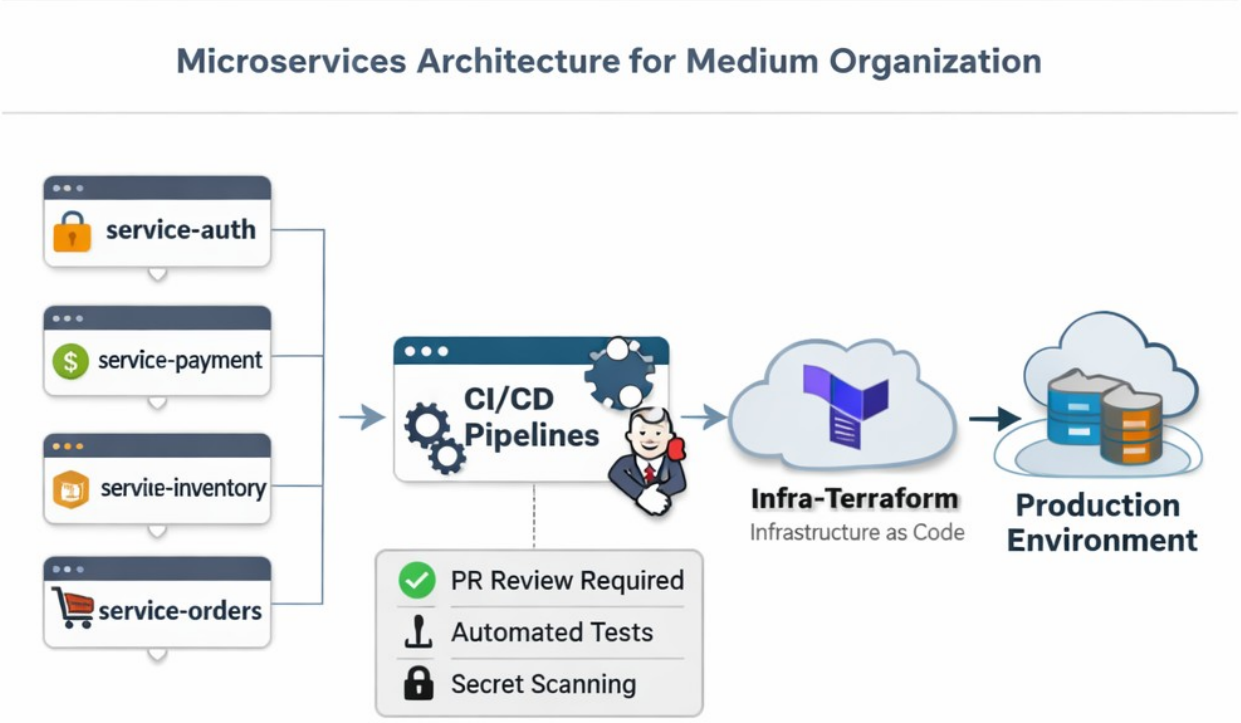
None yet

Notifications Customize

Workflow Explanation (short):

1. Clone repo → create feature branch → commit changes locally
2. Push feature branch to GitHub → open PR to main
3. Team reviews PR → merge → code is integrated into main safely
4. Optional: main branch protected to prevent direct pushes





COMPANY A — SMALL STARTUP

3 repos · trunk-based flow

backend-api
Node/Python API · REST endpoints
tests/ · .github/workflows/

frontend-ui
React/Vue SPA · components/
public/ · src/ · package.json

infra-devops
Docker · CI/CD pipelines
scripts/ · .env.example

Branch strategy

main → develop → feature/xxx

COMPANY B — MEDIUM ORG

6 repos · GitFlow + protection

svc-auth
Auth microservice

svc-orders
Orders microservice

svc-payments
Payments microservice

svc-notify
Notifications svc

infra-terraform
AWS / GCP IaC

platform-shared
Libs · API contracts

Branch strategy (GitFlow)

main → dev → feature → release → hotfix

🔒 main + dev protected

GitHub Policies Applied to Both

- **PR before merge** — No direct push to main
- **1+ reviewer required** — Approvals gate merge
- **Secret scanning on** — Block leaked credentials
- **Status checks pass** — CI must be green first
- **Signed commits** — Enforce GPG signing
- **CODEOWNERS file** — Auto-assign reviewers

Microservice Repo Internal Structure

svc-orders/ (example microservice layout)

src/
app.js / main.py
routes/
controllers/
models/
services/
utils/
config/
middleware/
index.js

tests/
unit/
integration/
e2e/
fixtures/
mocks/

deploy/
Dockerfile
docker-compose.yml
k8s/ (manifests)
helm/ (charts)
.env.example

.github/
workflows/
ci.yml
deploy.yml
CODEOWNERS
README.md
CHANGELOG
docs/
openapi.yml

Here's the full breakdown for both companies:

Task A — Company A: Small Startup Repo Design

Number of repos: 3

The key philosophy for a small startup is simplicity — no overhead, no bureaucracy. One repo per concern is enough.

`backend-api` holds all server-side code: the API logic, route handlers, database models, and any test files. `frontend-ui` contains the client app — React or Vue components, public assets, and the build config. `infra-devops` keeps everything operational: the Dockerfile, docker-compose files, CI pipeline configs, and environment templates.

Naming convention: `<purpose>-<technology>` in lowercase kebab-case. Short, readable, searchable.

Branching strategy — Trunk-Based (simplified):

`main` is always production-ready. Developers cut short-lived `feature/your-feature` branches and merge back via PR. A `develop` branch acts as a staging buffer before anything hits `main`. No long-lived branches, no release trains — just fast iteration.

Task B — Company B: Medium Organization Repo Design

Number of repos: 6

Each microservice gets its own isolated repo so teams can deploy independently, own their CI pipelines, and work without stepping on each other.

The four service repos — `svc-auth`, `svc-orders`, `svc-payments`, `svc-notify` — all follow the same internal folder structure (shown in the diagram above): `src/` for application code, `tests/` for unit/integration/e2e, `deploy/` for Docker and Kubernetes manifests, and `.github/` for CI workflows and CODEOWNERS.

`infra-terraform` is owned solely by the DevOps team and contains all cloud infrastructure as code — VPCs, RDS instances, IAM roles, load balancers — organized by environment (`envs/dev`, `envs/staging`, `envs/prod`).

`platform-shared` houses shared libraries, API contracts (OpenAPI specs), common utilities, and SDK packages that other services import.

Branch protection rules for `main` and `dev`:

- No force-push allowed
- Require 2 approvals for `main`, 1 for `dev`
- All CI status checks must pass before merge

- Branches must be up-to-date before merging (no stale PRs)

Task C — GitHub Policies (Minimum 3)

Policy 1 — Require PR before merge: Direct commits to `main` are blocked. Every change must go through a pull request, creating an audit trail and ensuring peer visibility.

Policy 2 — Require at least 1 reviewer: A PR cannot be merged by the author alone. For Company B, `main` requires 2 approvers; for Company A, 1 is sufficient. This prevents rubber-stamping.

Policy 3 — Secret scanning enabled: GitHub automatically scans all commits for hardcoded API keys, tokens, and credentials. Any match blocks the push and alerts the team — preventing the most common accidental exposure vulnerability.

Policy 4 — Required status checks: CI pipelines (lint, tests, build) must all pass green before a merge is permitted. No broken code enters the shared branches.

Policy 5 — CODEOWNERS enforcement: A `CODEOWNERS` file maps directories to responsible teams. Changing the `payments/` directory automatically requests review from the payments team — no one can sneak changes past the domain owners.

Policy 6 — Signed commits (Company B): GPG-signed commits verify that code actually came from the claimed developer — important in a multi-team environment with higher security requirements.